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CENG454 - Artificial Intelligence and Data Science
Final Project Document

Fake News Detection in Turkish News
Using Transformer-Based Sentence Embeddings

Prepared by: Group Members
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Project Summary

Project Topic: Fake news detection using transformer-based sentence embeddings.

Research Question: Can transformer-based sentence embeddings improve fake news detection performance compared to traditional TF-IDF-based machine learning methods?

Core Idea: News articles are represented using two different feature extraction approaches: TF-IDF and transformer-based sentence embeddings. These representations are classified as fake or real using machine learning classifiers such as Logistic Regression, Support Vector Machine, and Random Forest.

Expected Output: A reproducible experiment with preprocessing, training, evaluation metrics, confusion matrices, result tables, and a scientific report.

1. Introduction and Motivation

Fake news is a serious real-world problem because misleading or intentionally false information can affect public opinion, social trust, political decisions, economic behavior, and crisis response. With the rapid spread of online news and social media content, manually verifying every news article is not scalable.

Traditional text classification methods often represent text using word frequency-based approaches such as TF-IDF. Although these methods are simple and effective, they may fail to capture deeper semantic meaning, context, and sentence-level relationships. Transformer-based language models can generate dense sentence embeddings that represent semantic meaning more effectively.

This project investigates whether transformer-based sentence embeddings provide better performance than a traditional TF-IDF baseline for fake news detection. The project is designed as a data-driven AI experiment rather than only a software application.

2. Background and Related Work

Fake news detection is commonly treated as a binary text classification task where each news article is labeled as either real or fake. Common approaches include traditional machine learning methods, deep learning methods, and transformer-based models.

TF-IDF: A word-frequency-based representation that reflects how important a term is in a document relative to a corpus.

Logistic Regression: A simple and interpretable baseline classifier for binary classification problems.

Support Vector Machine: A strong traditional classifier that can perform well in high-dimensional text classification tasks.

Transformer-Based Embeddings: Dense semantic vector representations generated by transformer models such as BERT or Sentence-BERT.

Evaluation Metrics: Accuracy, precision, recall, F1-score, and confusion matrix are used to evaluate classification performance.

At least 10 recent and relevant academic sources should be included in the final bibliography. Suggested source categories include fake news detection surveys, transformer-based text classification papers, Sentence-BERT, Turkish NLP studies, and dataset papers or dataset cards.

3. Dataset and Preprocessing

The project can be implemented with either a Turkish fake news dataset or an English benchmark dataset. A Turkish dataset is preferable because it makes the project more locally relevant and original. However, an English dataset such as WELFake can be kept as a backup if the Turkish dataset has quality or access issues.

Option	Dataset	Advantage	Risk / Note
A	Turkish fake news dataset	More original and locally relevant	Dataset quality and label balance must be checked
B	WELFake / English dataset	Larger and easier to reproduce	Less original for Turkish context

Preprocessing Steps

- Remove rows with missing text or labels.
- Normalize label names into binary classes: fake and real.
- Convert text values to string type.
- Optionally lowercase text for TF-IDF experiments.
- Split the dataset into train and test sets using stratified sampling.
- Keep preprocessing steps reproducible with fixed random_state values.

4. Proposed Approach and Implementation

The project compares a traditional baseline approach with a transformer-based embedding approach. This makes the project scientifically stronger because the proposed method is evaluated against a clear baseline.

4.1 Experimental Pipeline

Raw News Dataset

- > Text Cleaning and Label Normalization
- > Train-Test Split
- > Feature Extraction
 - > TF-IDF Baseline
 - > Transformer Sentence Embeddings
- > Classifier Training
 - > Logistic Regression
 - > Support Vector Machine
 - > Random Forest (optional)
- > Prediction
- > Evaluation and Error Analysis

4.2 Baseline Method

The baseline method uses TF-IDF vectorization followed by Logistic Regression. This baseline is simple, fast, and suitable for comparison with transformer embeddings.

```

from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.pipeline import Pipeline

tfidf_model = Pipeline([
    ("tfidf", TfidfVectorizer(max_features=10000)),
    ("clf", LogisticRegression(max_iter=1000))
])

tfidf_model.fit(X_train, y_train)

```

4.3 Transformer Embedding Method

The main approach uses a SentenceTransformer model to convert each news article into a dense vector representation. These embeddings are then classified with traditional machine learning classifiers.

```

from sentence_transformers import SentenceTransformer
from sklearn.linear_model import LogisticRegression

embedding_model = SentenceTransformer(
    "sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2"
)

X_train_emb = embedding_model.encode(X_train.tolist(), show_progress_bar=True)
X_test_emb = embedding_model.encode(X_test.tolist(), show_progress_bar=True)

clf = LogisticRegression(max_iter=1000)
clf.fit(X_train_emb, y_train)

```

5. Evaluation and Results

The final report should include quantitative results for each method. The main metrics are accuracy, precision, recall, F1-score, and confusion matrix. Since class imbalance may occur in fake news datasets, F1-score is especially important.

Method	Classifier	Accuracy	Precision	Recall	F1-score
TF-IDF	Logistic Regression	0.xx	0.xx	0.xx	0.xx
Transformer Embedding	Logistic Regression	0.xx	0.xx	0.xx	0.xx
Transformer Embedding	SVM	0.xx	0.xx	0.xx	0.xx
Transformer Embedding	Random Forest (optional)	0.xx	0.xx	0.xx	0.xx

The numeric values above must be replaced with real experimental results. Results should not be invented. The final report should also include at least one confusion matrix figure and a short error analysis of wrongly classified examples.

```

from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

y_pred = clf.predict(X_test_emb)

```

```
print("Accuracy:", accuracy_score(y_test, y_pred))
print(classification_report(y_test, y_pred))
print(confusion_matrix(y_test, y_pred))
```

6. Discussion and Limitations

The discussion section should explain not only which model performed best, but also why it may have performed better or worse. For example, transformer embeddings may capture semantic similarity better than TF-IDF, but TF-IDF may still perform strongly when fake news contains repeated keywords or stylistic patterns.

Possible limitations include dataset bias, class imbalance, source-specific writing styles, limited generalization to unseen topics, and the difficulty of detecting misinformation that is factually false but linguistically similar to real news.

7. Use of Artificial Intelligence Tools

This project contains two different types of AI usage. The first one is methodological AI usage inside the experiment. The second one is external AI-assistance usage during planning, writing, and debugging.

7.1 AI Used Inside the Project

Artificial intelligence is used as part of the fake news detection pipeline. A transformer-based sentence embedding model converts news articles into dense numerical vector representations. These vectors are then used by machine learning classifiers to predict whether a news article is fake or real.

7.2 AI Tools Used During Project Development

The following statement can be included in the final report:

AI Tool Usage Statement: During this project, ChatGPT was used as an assistant for brainstorming the project structure, improving the clarity of explanations, and debugging selected code snippets. It was not used to generate the final results automatically. All experiments, dataset preprocessing, model training, evaluation, and interpretation of results were performed and verified by the project members. Any AI-assisted text was reviewed, modified, and validated by the authors.

This statement is important because the course instructions require any use of AI tools such as ChatGPT, Copilot, or Gemini to be explicitly documented. Excessive reliance on generative tools without critical analysis should be avoided.

8. Reproducibility and Submission Package

The project should be submitted as a single zip file including the final PDF report, source code files, and the dataset or dataset links. The following structure is recommended:

```
CENG454_Group_X/
|-- report/
|   |-- final_report.pdf
```

```
|-- notebooks/  
|   |-- fake_news_detection.ipynb  
|-- src/  
|   |-- preprocessing.py  
|   |-- train_tfidf.py  
|   |-- train_embeddings.py  
|   |-- evaluate.py  
|-- results/  
|   |-- confusion_matrix_tfidf.png  
|   |-- confusion_matrix_transformer.png  
|   |-- results_table.csv  
|-- data/  
|   |-- README_dataset_link.txt  
|-- README.md
```

9. Conclusion and Future Work

This project aims to evaluate transformer-based sentence embeddings for fake news detection and compare them with a traditional TF-IDF baseline. The expected contribution is a reproducible experimental comparison showing whether semantic embeddings improve fake news classification performance.

Future work may include fine-tuning a Turkish BERT model, testing multilingual datasets, applying zero-shot LLM classification, adding explainability methods, and analyzing whether models generalize across different news topics and sources.

10. References to Add in IEEE Format

The final report should include at least 10 relevant and recent references in IEEE format. Suggested references include:

1. Sentence-BERT paper
2. BERT paper
3. Fake news detection survey papers
4. Transformer-based text classification papers
5. Turkish NLP / BERTurk papers
6. Dataset source citation or dataset card
7. Scikit-learn documentation for models and metrics
8. SentenceTransformers documentation
9. Recent misinformation detection papers
10. Ethical AI / misinformation analysis papers